

PAPER_693: Sombrero Galaxy (M104/NGC 4594): Edge-On Sa Dynamics and UQFF

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Abstract

The Sombrero Galaxy (M104/NGC 4594) is a nearby (8.6 Mpc) nearly edge-on Sa galaxy with an exceptionally massive bulge, prominent dust ring, and a $10^9 M_\odot$ central black hole. Its dynamics are modeled via UQFF-modified gravitational potentials.

1. System Parameters

Parameter	Value
M_{bulge}	$8 \times 10^{11} M_\odot$
M_{disk}	$2 \times 10^{11} M_\odot$
M_{BH}	$10^9 M_\odot$
R_{eff}	4.2 kpc
Inclination	84°

2. Circular Velocity

$$v_c(R) = \sqrt{\frac{GM_{enc}(R)}{R}} \left(1 + \frac{\rho_{SCm}}{2\rho_{UA}} \right)$$

3. Gravitational Potential (UQFF-modified)

$$\Phi_{UQFF}(r) = -\frac{GM_{bulge}}{r+a} - \frac{GM_{disk}}{2R_d} \ln(R^2 + h^2) + \frac{\Lambda c^2 r^2}{6}$$

where $a = R_{eff}$ is the Hernquist scale radius.

4. Bulge Velocity Dispersion

$$\sigma_{bulge} = \sqrt{\frac{GM_{bulge}}{5R_{eff}}} \approx 230 \text{ km/s}$$

5. Dust Lane Wavefunction

$$\Psi_{dust}(z, t) = e^{-z^2/(2h^2)} \cos(\omega_i t)$$

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Kormendy et al. (1996), ApJ, 473, L91
- UQFF Framework, Star-Magic Session 174

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